

The core of adjoint functors 29/11/2011 ^①

Kan [1958] introduced the notion of adjoint functors. By defining the unit and counit natural transformations, he paved the way for the notion to be internalized to any 2-category. This was done by Kelly [1969] whose interest was particularly in the 2-category of V -categories for a monoidal category V (in the sense of Eilenberg-Kelly [1965]).

For V -categories A and X , an adjunction consists of

A1. V -functors $U : A \rightarrow X$ and $F : X \rightarrow A$;

A2. a family of isomorphisms

$$\kappa : A(FX, A) \cong X(X, UA)$$

in V which is V -natural in

$A \in A$ and $X \in X$.

We write $\kappa : F \dashv U : A \rightarrow X$.

The following result is well known [Kelly Book; §1.11].

Proposition 1 Suppose $U: A \rightarrow X$ is a V -functor, $F: \text{ob } X \rightarrow \text{ob } A$ is a function, and $\pi: A(FX, A) \cong X(X, UA)$ is a family of isomorphisms in $\mathcal{V}$ which are V -natural in $A \in A$ for each $X \in X$. Then there exists a unique adjunction $\pi: F \dashv U: A \rightarrow X$ for which $F: \text{ob } X \rightarrow \text{ob } A$ is the effect of the V -functor $F: X \rightarrow A$ on objects.

Definition For V -categories A and X , an adjunction core consists of

- c1. functions $U: \text{ob } A \rightarrow \text{ob } X$ and $F: \text{ob } X \rightarrow \text{ob } A$;
- c2. a family of isomorphisms

$$\pi: A(FX, A) \cong X(X, UA), \quad A \in A, X \in X, \text{ in } \mathcal{V}.$$

Given such a core, we make the following definitions:

(3)

(i) $\eta_X : X \rightarrow UFX$ is the composite

$$I \xrightarrow{j} A(FX, FX) \xrightarrow{\pi} X(X, UFX);$$

(ii) $\varepsilon_A : FUA \rightarrow A$ is the composite

$$I \xrightarrow{j} X(UA, UA) \xrightarrow{\pi^{-1}} A(FUA, A);$$

(iii) $U_{AB} : A(A, B) \rightarrow X(UA, UB)$ is the composite

$$A(A, B) \xrightarrow{A(\varepsilon_A, 1)} A(FUA, B) \xrightarrow{\pi} X(UA, UB);$$

(iv) $F_{XY} : X(X, Y) \rightarrow A(FX, FY)$ is the composite

$$X(X, Y) \xrightarrow{X(1, \eta_Y)} X(X, UFY) \xrightarrow{\pi^{-1}} A(FX, FY).$$

Clearly each adjunction $\pi : F \dashv U : A \rightarrow X$ gives rise to an adjunction core. It is well known then that the effect of U and F on homs are as in (iii) and (iv).

Theorem 2 An adjunction core extends to an adjunction if and only if one of the diagrams (v) or (vi) commutes. The adjunction is unique when it exists.

$$\begin{array}{ccc}
 A(A, B) \otimes A(FX, A) & \xrightarrow{u_{AB} \otimes \pi} & X(UA, UB) \otimes X(X, UA) \\
 \text{comp} \downarrow & & \downarrow \text{comp} \\
 A(FX, B) & \xrightarrow{\pi} & X(X, UB)
 \end{array}$$

(v)

$$\begin{array}{ccc}
 X(Y, UA) \otimes X(X, Y) & \xrightarrow{\pi^{-1} \otimes F_{X, Y}} & A(FY, A) \otimes A(FX, FY) \\
 \text{comp} \downarrow & & \downarrow \text{comp} \\
 X(X, UA) & \xrightarrow{\pi^{-1}} & A(FX, A)
 \end{array}$$

(vi)

Proof We deal first with the version involving (v). For an adjunction, (v) expresses the V -naturality of π in $A \in \mathcal{A}$. Conversely, given an adjunction core satisfying (v), we paste to the left of (v) with $X = UC$, the diagram

$$\begin{array}{ccc}
 A(A, B) \otimes A(C, A) & \xrightarrow{1 \otimes A(\varepsilon_C, 1)} & A(A, B) \otimes A(FUC, A) \\
 \text{comp} \downarrow & & \downarrow \text{comp} \\
 A(C, B) & \xrightarrow{A(\varepsilon_C, 1)} & A(FUC, B)
 \end{array}$$

which commutes by naturality of composition. This leads to the commutative square

(5)

$$\begin{array}{ccc}
 A(A, B) \otimes A(C, A) & \xrightarrow{U \otimes U} & X(UA, UB) \otimes X(UC, UA) \\
 \text{comp} \downarrow & & \downarrow \text{comp} \\
 A(C, B) & \xrightarrow{U} & X(UC, UB)
 \end{array}$$

which shows that U preserves composition. We also have commutativity of

$$\begin{array}{ccc}
 I & \xrightarrow{j} & A(A, A) \\
 \downarrow \varepsilon_A & \searrow A(\varepsilon_A, 1) & \downarrow U \\
 A(FUA, A) & \xrightarrow{\pi} & X(UA, UB) \\
 \downarrow j & \swarrow \pi & \downarrow U \\
 & & X(UA, UB)
 \end{array}$$

(ii) (iii)

completing the proof that U is a $\mathcal{V}$ -functor. Now the general (v) expresses the $\mathcal{V}$ -naturality of π in A . By Proposition 1, we have an adjunction determined uniquely by the core.

If we write $\mathcal{V}^{\text{rev}}$ for $\mathcal{V}$ with reversed monoidal structure $A \otimes^{\text{rev}} B = B \otimes A$, and apply the first part of the proof to the $\mathcal{V}^{\text{rev}}$ -enriched adjunction $\pi^!: U^{\text{op}} \dashv F^{\text{op}}: X^{\text{op}} \rightarrow A^{\text{op}}$, we see it is equivalent to an adjunction core satisfying (vi). $\square$

Corollary 3 If $\mathcal{V}$ is a poset then adjunction cores are adjunctions.

Proof All diagrams, including (v), commute in $\mathcal{V}$. $\square$